Simulating the Price Values of TPL Stock for the Next 20 Days Using Geometric Brownian Motion

a) Created the time index

The time index value corresponding to the first price value of TPL stock in the data file as t = 0, second price values of TPL is t=1 etc.

Then range of t is [0,479]

	Date	Closing Price	Time_index
0	10/3/2022	98.80	0
1	10/4/2022	94.50	1
2	10/5/2022	98.70	2
3	10/6/2022	98.50	3
4	10/7/2022	100.00	4
...	...	...	...
475	9/25/2024	124.75	475
476	9/26/2024	122.00	476
477	9/27/2024	126.50	477
478	9/30/2024	126.25	478
479	10/1/2024	126.25	479

b) Suppose that price values follow Geomatric Brownian motion with drift parameter μ and scale parameter σ then theoretically we can prove the distribution of log return like below,

$X_t = exp\left((\mu - \frac{\sigma^2}{2})t + \sigma W_t\right) \sim \log normal$ $\quad where\ \{W_t; t \geq 0\}$ is standard Brownian motion.

$$\ln(X_t) = \left((\mu - \frac{\sigma^2}{2})t + \sigma W_t\right) \sim N(\left(\mu - \frac{\sigma^2}{2}\right)t, \sigma^2 t)$$

$$\ln(X_{t-1}) = \left(\left(\mu - \frac{\sigma^2}{2}\right)(t-1) + \sigma W_{t-1}\right) \sim N(\left(\mu - \frac{\sigma^2}{2}\right)(t-1), \sigma^2(t-1))$$

Then,

$$\boldsymbol{LogReturn_t = \ln(X_t) - \ln(X_{t-1}) = \left(\left(\mu - \frac{\sigma^2}{2}\right) + \sigma(W_t - W_{t-1})\right) \sim N(\left(\mu - \frac{\sigma^2}{2}\right), \sigma^2)}$$

Then the Log Returns are identically and normally distributed with mean $\left(\mu - \frac{\sigma^2}{2}\right)$ $and\ variance\ \sigma^2$.

$$\boldsymbol{\ln(X_t) - \ln(X_{t-1}) \sim N(\left(\mu - \frac{\sigma^2}{2}\right), \sigma^2)}$$

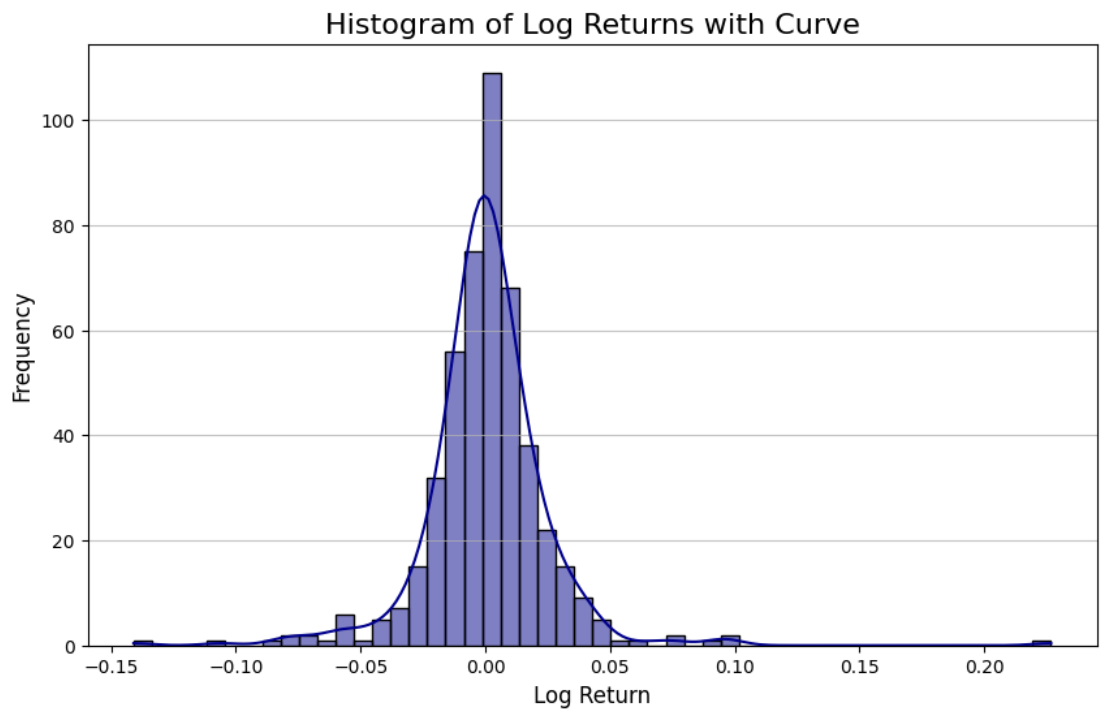


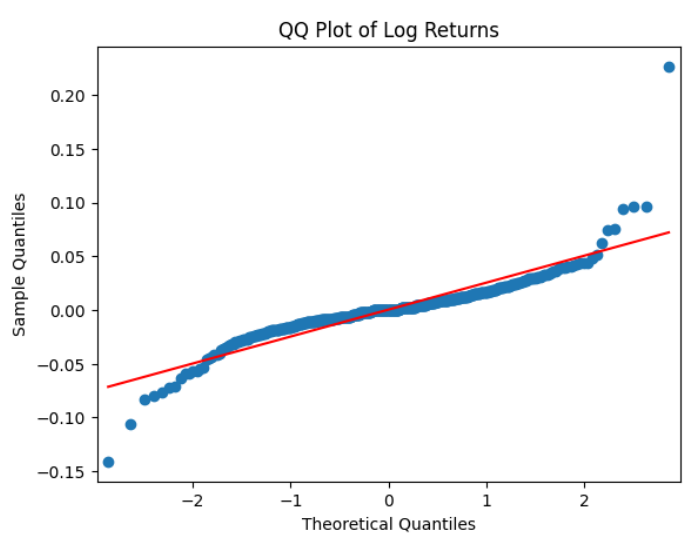
Figure 1: Log return Price values of TPL

c) Parameter Estimation

drift parameter is 0.**0008268** by taking means of log return values.
Scale parameter is **0.02510** by taking standard deviation of log return values.

d) Is this model is appropriate?

- Check the normality by Q-Q plot



The Q-Q plot of log returns indicates a deviation from normality. In an ideal normal distribution, the points would closely follow the red reference line. However, in this plot, some points deviate noticeably from the line, particularly in the tails — both on the lower and upper ends. But most of the plotted points lies near the theoretical fitted line.

- Test the Normality by Shapiro Wilk test

Results of the test

```
Statistics=0.837, p=0.000
Reject h0, the distribution is not normal
```

The Shapiro-Wilk test for normality yielded a statistically significant result ($p < 0.05$), leading to the rejection of the null hypothesis of normality. This suggests that while the distribution appears roughly normal upon visual inspection, it does not meet the strict statistical criteria for normality.
so, we can conclude that the log returns are approximately normally distributed, with some deviations.

- Check the Independence of Log returns
Applying hypothesis test the Ljung-Box test on the raw log-returns, the test results gives p values as 0.004086 less than 0.05 which rejects the null hypothesis of no autocorrelation.

```
Ljung-Box Test Results for Raw Log-Returns:
      lb_stat  lb_pvalue
10  25.753815   0.004086
```

Since GBM requires both **normally distributed** and **independent increments**, the violations of these assumptions indicate that this model is **not fully appropriate** for simulating GBM.

e) Simulate 10,000 possible sample paths for the closing price of TPL relevant to the next 20 days

Paths are simulated using BrownianG function used in reference videos.

Parameters are:

```
y=BrownianG(10000,20,20,sigma,mu,w0,GB=True)
```

- ✓ No of sample paths = 10,000
- ✓ Total time = 20 days
- ✓ No of time steps = 20
- ✓ Sigma and drift parameters
- ✓ wo is closing price at t=479 which is 126.25

	0	1	2	3	4	5	6	7	8	9	...	9990	9991	9992	9993	9994	9995	9996	9997	9998	9999
0	126.250000	126.250000	126.250000	126.250000	126.250000	126.250000	126.250000	126.250000	126.250000	126.250000	...	126.250000	126.250000	126.250000	126.250000	126.250000	126.250000	126.250000	126.250000	126.250000	126.250000
1	124.082307	126.535765	129.297905	126.570619	124.195717	127.527158	125.976305	129.903458	127.415116	124.787763	...	126.067710	122.321495	127.297760	131.132649	130.878623	125.676104	125.647255	123.912710	122.567390	131.058109
2	124.840594	127.468501	136.740758	129.699710	126.293801	131.963083	124.503264	127.375683	123.313378	129.210553	...	127.784027	120.789977	123.952069	133.660716	131.201017	125.505176	121.672924	120.168493	123.744526	139.409572
3	127.556823	129.226871	134.869101	127.315749	124.140923	134.900212	128.643054	125.056035	128.764151	132.107352	...	127.167251	129.464162	123.248891	136.195096	134.182258	123.565626	124.770702	124.728424	124.238309	142.042544
4	125.689750	126.948494	130.455133	125.501570	125.529142	135.657991	132.803342	126.889258	130.155008	133.730882	...	125.273129	130.244123	121.961430	131.966206	136.730501	123.340282	120.763480	125.102399	123.247567	148.026700
5	130.174064	130.420156	129.424423	120.595634	123.630558	136.381684	127.793381	124.741443	129.072880	124.719691	...	127.386457	130.111792	118.868860	129.070596	143.104457	124.901409	121.480559	124.203883	122.181243	148.625071
6	129.154884	135.347631	126.405557	119.670211	125.553679	133.728695	126.393689	127.066628	130.695220	122.131639	...	126.320356	132.142473	117.526520	129.098734	145.901356	126.039205	124.682086	131.623534	121.260554	146.361518
7	129.659621	134.432921	122.944448	122.533060	130.926801	132.658466	127.177427	130.016650	135.446831	124.192349	...	129.461982	127.291274	123.178966	141.624972	141.955731	118.141055	131.115429	135.443222	123.527888	152.925983
8	128.584000	130.090227	126.199100	121.538802	133.162423	131.015586	119.224507	127.781225	132.958191	127.653154	...	126.568192	126.144388	129.009111	144.840468	144.192903	118.473546	131.557204	137.889388	124.857209	153.009301
9	131.995874	131.011497	128.452639	122.167827	129.590861	125.611995	118.361304	132.405076	134.572426	124.232901	...	129.026777	122.764005	129.798585	141.498657	143.578203	116.641550	136.876169	132.138523	126.507154	149.838444
10	137.480701	132.814659	130.506233	115.445244	135.043658	121.287030	115.767699	137.150524	135.476716	122.018451	...	122.832377	127.904988	126.041100	147.683538	151.778597	113.866864	137.501860	130.389191	130.408482	154.599828
11	137.898320	129.728081	128.817447	111.834660	132.163334	119.691186	118.758229	135.731989	137.009046	119.719295	...	124.903156	129.446455	130.680497	153.391363	147.727222	112.121737	136.953028	127.857523	133.554862	160.985468
12	136.454999	127.475378	128.122917	111.927064	132.440199	121.069961	114.489287	137.765566	133.024469	123.172101	...	119.861561	124.599832	128.074199	153.349861	149.569030	113.264457	142.730115	129.590228	130.448345	159.519023
13	134.610850	127.048815	124.915438	111.560978	136.918246	118.787181	114.005991	136.998397	135.277890	122.681902	...	117.542398	127.325602	125.209967	158.177850	148.453722	114.454734	145.061585	132.647954	127.834782	161.648144
14	131.921778	132.034593	130.859000	113.703257	133.783239	118.687478	121.588034	135.779822	130.511622	125.856841	...	122.051185	127.237493	123.787902	154.812883	147.643058	112.148336	140.626316	125.867173	128.836113	157.930436
15	129.540343	126.202584	127.986849	116.702079	130.494911	120.433501	116.460316	130.818299	126.201872	126.495071	...	120.546548	130.396648	128.471472	150.917851	144.257215	110.392248	145.207306	125.356948	127.207027	162.151513
16	130.433334	128.539366	132.649836	114.754407	134.599916	118.504985	113.773937	129.574849	125.948379	130.804475	...	119.151506	131.899462	128.720538	154.798045	135.421869	113.424410	143.680243	126.787516	124.681105	163.528184
17	129.481844	127.946126	128.435893	112.933994	138.934705	120.639811	117.195343	129.458326	127.090015	133.076404	...	124.710000	130.799580	128.101238	151.879828	134.039379	116.263008	144.093780	128.081768	124.567079	167.106807
18	140.716898	131.478302	128.504210	109.674490	138.521937	118.823902	117.975362	133.541679	125.894101	132.016010	...	127.502845	127.506784	125.872731	149.023067	136.898577	117.871558	144.234320	125.847138	124.780735	168.734661
19	141.669471	130.232357	128.410786	111.250199	139.645019	117.210170	117.873338	131.926139	126.692255	130.530059	...	128.998786	125.751846	121.733542	147.507710	143.055849	114.623319	142.122212	132.022953	119.266214	167.091957
20	138.629978	129.037927	130.327300	114.161790	136.731461	115.600354	114.149119	134.667284	133.239361	127.894991	...	128.988373	121.938495	118.498274	141.235930	140.757799	112.859123	140.253275	126.128338	115.512332	173.616875

21 rows × 10000 columns

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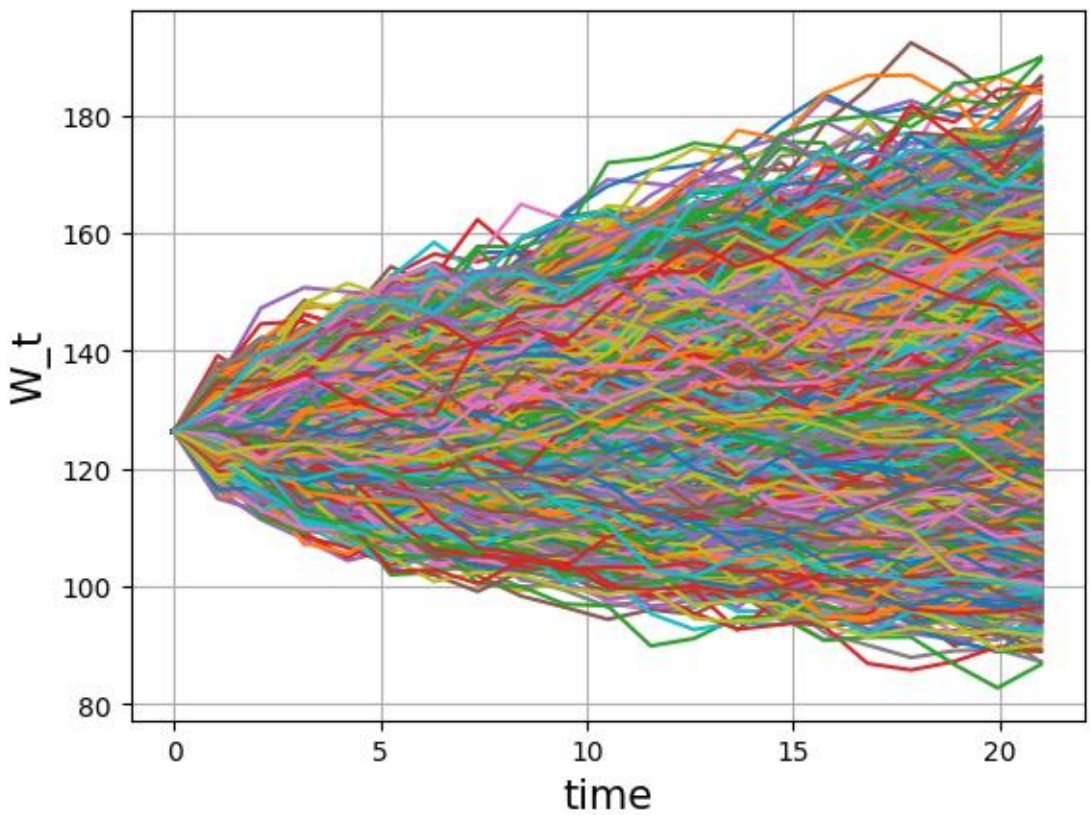
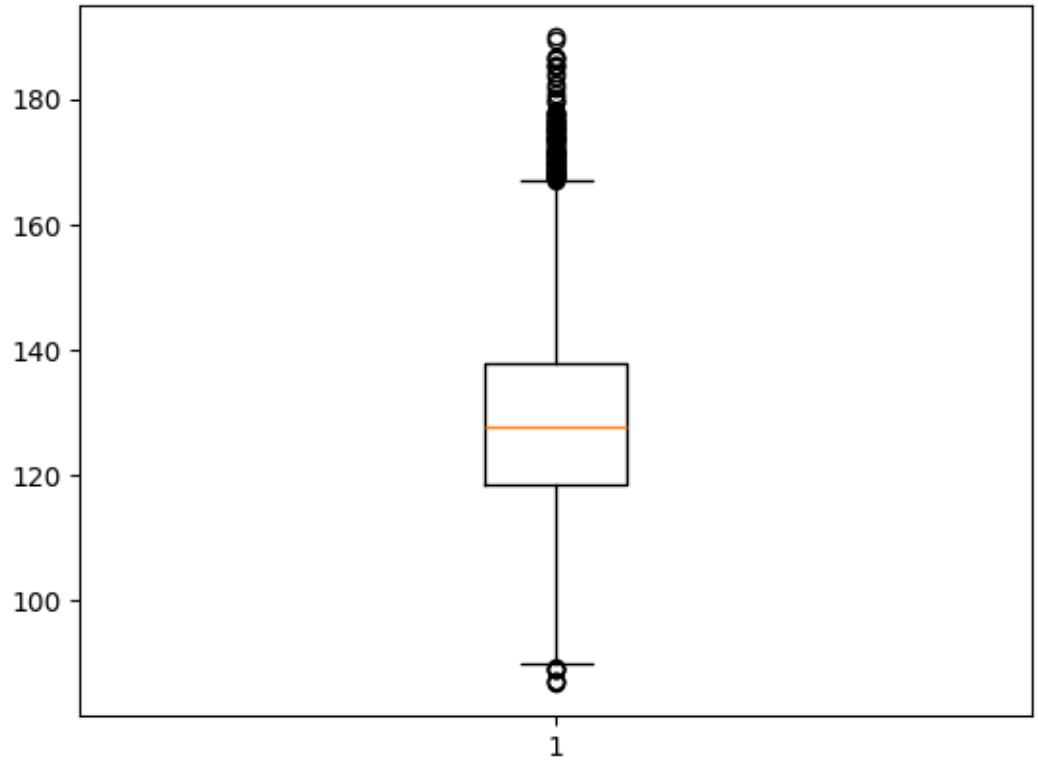
Note: here one column represent the one sample path.

f) Plot the sample paths for the next 20 days.



g) The box plot for the possible closing price values of TPL corresponding to the last day of the next 20-day period.



h) What is the probability that the closing price corresponding to the last day of the next 20-day period is between 130 and 140?
Need to get values at last day of next 20 days which lies between 130 and 140 and divided by total no of values.
P(130<prices<140) = Number of closing prices in between 130 and 140 / total number of pries
= **0.2243**

i) The 95% prediction interval for the closing price value corresponding to the last day of the next 20-day period
[102.47631089, 159.1395269]

Python Code for the simulation:

https://github.com/thamodya2001/GBM-simulation-